

END SEMESTER EXAMINATIONS, AUTUMN 2025

Program: BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & ENGINEERING)			
Subject Name: INTRODUCTION TO DATA SCIENCE			
Subject Code: 2000016103			Semester: VII
Duration: 2 Hours		Maximum Marks: 50	

Note: Answer the following questions as directed.

Mod. No.- Module/Topic/Unit No., **BL-** Blooms Taxonomy, **CO-** Course Outcome.

Group A
(Answer All questions)
(Each question carries 1 mark)

1. Write the correct answer from the options given: (10 X 1)

		Mod. No.	BL	CO
(i)	Data Science is concerned with: (I) Handling Large Volume of data. (II) It is an interdisciplinary subject. (III) Data Science is an important branch of AI.	1	1	1
	(a) Only (I) is correct. (b) (I) and (II) is correct. (c) (I), (II) and (III) is (d) None of the mentioned is correct. correct.			
(ii)	By 'Peta byte' we mean	1	1	1
	(a) 1×10^6 bytes (b) 1×10^{12} bytes (c) 1×10^{15} bytes (d) 1×10^{24} bytes			
(iii)	In Uni-variate data analysis the number of independent variable is :	1	2	2
	(a) one (b) Two (c) Many (d) None of the mentioned			
(iv)	To measure the average value of a set of data, (I) Mean is the most faithful measurement. (II) Median is the most faithful measurement.	1	2	2
	(a) (I) is correct. (b) (II) is correct. (c) (I) and (II) both are (d) Neither (I) nor (II) is correct. correct.			

		Mod. No.	BL	CO
(v)	One of the main purpose of linear regression is:	1	2	2
	(a) Estimating the value of a random variable.			
	(b) Estimating the value of one variable from the other.			
	(c) Both are correct.			
	(d) None of them are correct.			
(vi)	The appropriate regression method used to classify data is:	1	2	2
	(a) Linear regression.			
	(b) Logistic regression.			
	(c) Nonlinear regression.			
	(d) None of the mentioned			
(vii)	What is perceptron?	2	1	3
	(a) A single layer feed forward neural network with preprocessing.			
	(b) A double layer auto associative neural network.			
	(c) An auto associative neural network.			
	(d) None of the mentioned			
(viii)	Which one of the following is true for neural network	2	1	3
	(a) Each node computes it weighted input.			
	(b) Node could be in excited or non-excited mode.			
	(c) It has set of nodes and connections.			
	(d) All of the mentioned			
(ix)	For noisy data the value of 'k' in k-n-n algorithm must be:	2	2	3
	(a) Small			
	(b) Large			
	(c) Medium			
	(d) All of the mentioned			
(x)	The main computational framework for Big Data processing is:	3	1	5
	(a) Map-Reduce			
	(b) HDLC			
	(c) Both of them.			
	(d) None of them.			

Group B

(Answer any Four questions out of Six)
(Each question carries 5 marks)

(4 X 5)

2. Prove that in linear regression the least square line always passes through the average values of the independent and dependent data.

Mod. No.	BL	CO
1	5	2

Mod. No.	BL	CO
1	3	2
2	3	3
2	5	3

3. If the random variable is discrete in nature then out of PMF and PDF which gives the correct distribution function?

The PMF of a discrete random variable is given by:

$$P(x) = \frac{1}{2^x} \text{ if } x \geq 0, \\ = 0 \text{ otherwise.}$$

Calculate $P(x \in A)$ if $A = \text{set of odd integers}$.

4. An artificial neural network is described with the following parameters:

- No. of input nodes = 3.
- No. of output node = 1.
- The input signals are given by $[x_1 = 0.8, x_2 = 0.6 \text{ and } x_3 = 0.4]$.
- The weights are given by $[w_1 = 0.1, w_2 = 0.3 \text{ and } w_3 = -0.2]$.
- The bias $b = 0.35$ with weight = 1.
- The pre-activation function is $z = \sum w_i x_i + b$
- The activation function of the output node is:
- $f(x) = \frac{1}{1+e^{-z}} - 1$.

Draw the schematic diagram of the network and calculate the output.

5. Draw a neural network with two input nodes and one output node. The weights are $W_1 = 1, W_2 = 1$. Describe with justification what will be the activation function to realize 'OR' logic using the network. Draw the input in a rectangular coordinate system and draw the class boundary to separate the True output from the False output.

6. A multilayer neural network has two input node, one hidden layer with two nodes and two output node. Draw the architecture of the network and show that each node of the hidden layer draws a straight line on the feature plane.
7. Design a decision rule to test the hypothesis that a coin is fair if a sample of 64 tosses of the coin is considered and if the level of significance is 0.005. Given $z=1.96$ corresponds to the area under the normal curve = 0.4750.

Mod. No.	BL	CO
2	4	3
1	6	2

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. Derive the expression of the delta rule for a single neuron using back propagation and gradient descent.
9. Techmart is an online service provider. They conducted a survey and claims that individual customers pay on average Rs. 600 per month with a standard deviation of Rs. 25. A random sample of 50 Customer's bill during a given month is taken with a mean of Rs. 550 and standard deviation Rs. 15.
- (a) Identify the null hypothesis and the alternate hypothesis.
- (b) Justify with proper explanation that the null hypothesis is accepted or rejected.

Mod. No.	BL	CO
2	4	3
1	3	2

10. Describe K-N-N algorithm and give logic how to select the K values for a given problem.

11. Describe briefly the steps of the Map-reduce algorithm to multiply two matrices as given below:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 5 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 1 & 5 & 1 \\ 5 & 1 & 3 & 4 \end{bmatrix}$$

Mod. No.	BL	CO
2	1	3
3	3	5